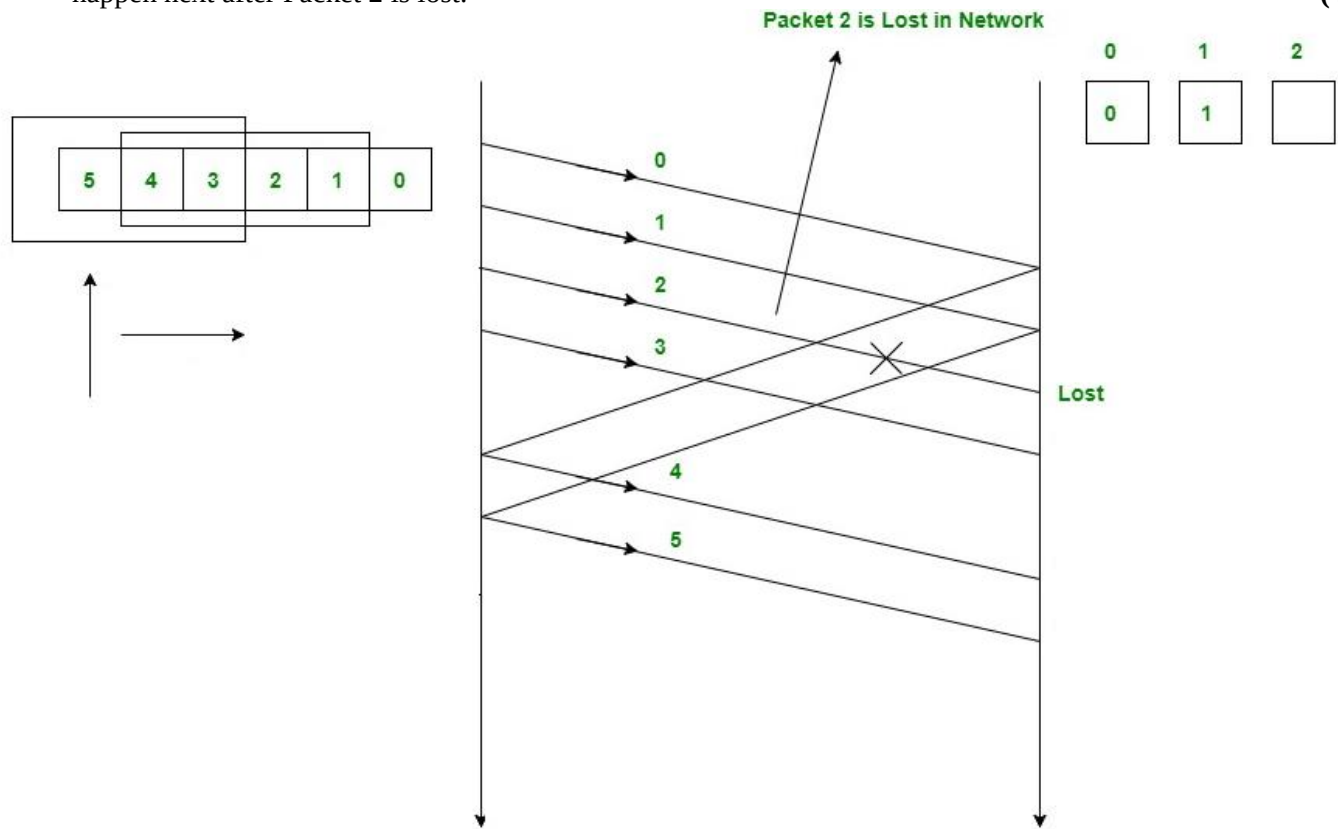


- a. The sender is at the ready state and an error-free ACK arrives. →
- b. The sender is at the blocking state and a time-out occurs. →
- c. The sender is at the ready state and a time-out occurs.→

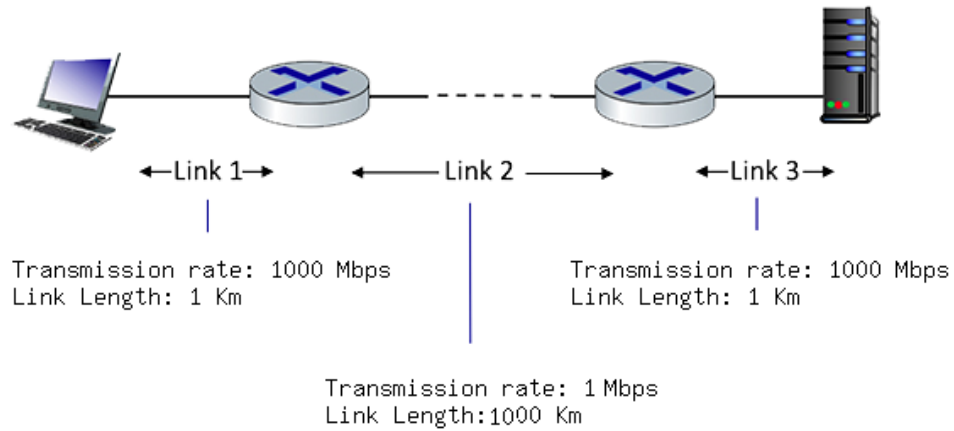
5. Answer the following questions related to the GBN protocol. Label the diagram properly and also show what will happen next after Packet 2 is lost. (4)



6. Fill the right side with the appropriate layer of the TCP/IP Model, hub, switch, router device. (6)

Description	Device or TCP/IP Layer
This device sends and receives information about the Network layer.	
This layer uses service access points.	
This device uses hardware addresses to filter a network.	
Ethernet is defined at these layers. (2 layers)	
This layer supports flow control and sequencing.	
This device can measure the distance to a remote network.	
Logical addressing is used at this layer.	
Hardware addresses are defined at this layer.	
This device creates one big collision domain and one large broadcast domain.	
This device creates many smaller collision domains, but the network is still one large broadcast domain.	
This device can never run full duplex.	
This device breaks up collision domains and Broadcast domains	

7. Consider the figure below, with three links, each with the specified transmission rate and link length.



Assume the length of a packet is 16000 bits. The speed of light propagation delay on each link is 3×10^8 m/sec. Calculate the total delay? **(4)**

8. Is Netflix multicast or unicast? Justify your answer. **(4)**

9. Is it possible for an organization's Web server and mail server to have exactly the same alias for a hostname (for example, Google.com)? What would be the type for the RR that contains the hostname of the mail server? **(4)**

10. CDNs typically adopt one of two different server placement philosophies. Name and briefly describe them. **(4)**

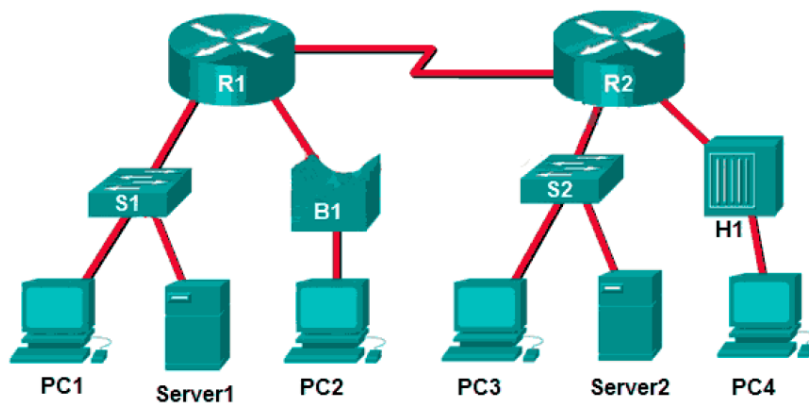
11. Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (5)

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host: gai
a.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (
Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gec
ko/20040804 Netscape/7.2 (ax) <cr><lf>Accept:ex
t/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5
<cr><lf>Accept-Language: en-us, en;q=0.5<cr><lf>Accept-
Encoding: zip, deflate<cr><lf>Accept-Charset: ISO
-8859-1, utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive: 300<cr>
<lf>Connection:keep-alive<cr><lf><cr><lf>
```

- What is the URL of the document requested by the browser?
- What version of HTTP is the browser running?
- Does the browser request a non-persistent or a persistent connection?
- What is the IP address of the host on which the browser is running?
- What type of browser initiates this message? Why is the browser type needed in an HTTP request message

For Farhan Siddiqui's Sections

12. How many broadcast and collision domains exist in the topology? (4)



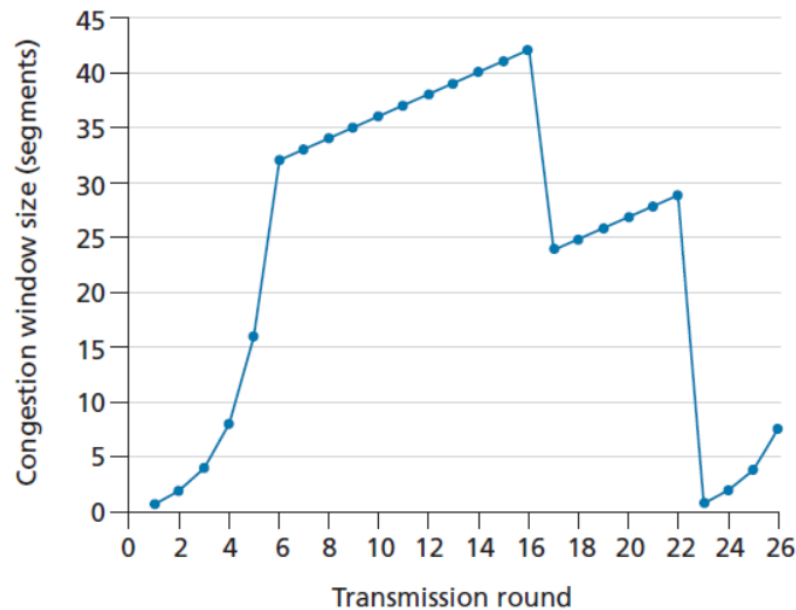
Collision Domain _____

Broadcast Domain _____

For Waseem Arain's Sections

12. Consider the following figure. Assuming TCP Reno is the protocol, answer the following questions. In all cases, you should provide a short reason justifying your answer. **(14)**

- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating.



- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- What is the initial value of ssthresh at the first transmission round?
- What is the value of ssthresh at the 18th transmission round?
- What is the value of ssthresh at the 24th transmission round?
- During what transmission round is the 70th segment sent?
- Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?
- Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?
- Again suppose TCP Tahoe is used, and there is a timeout event at 22nd round. How many packets have been sent out from 17th round till 22nd round, inclusive?